

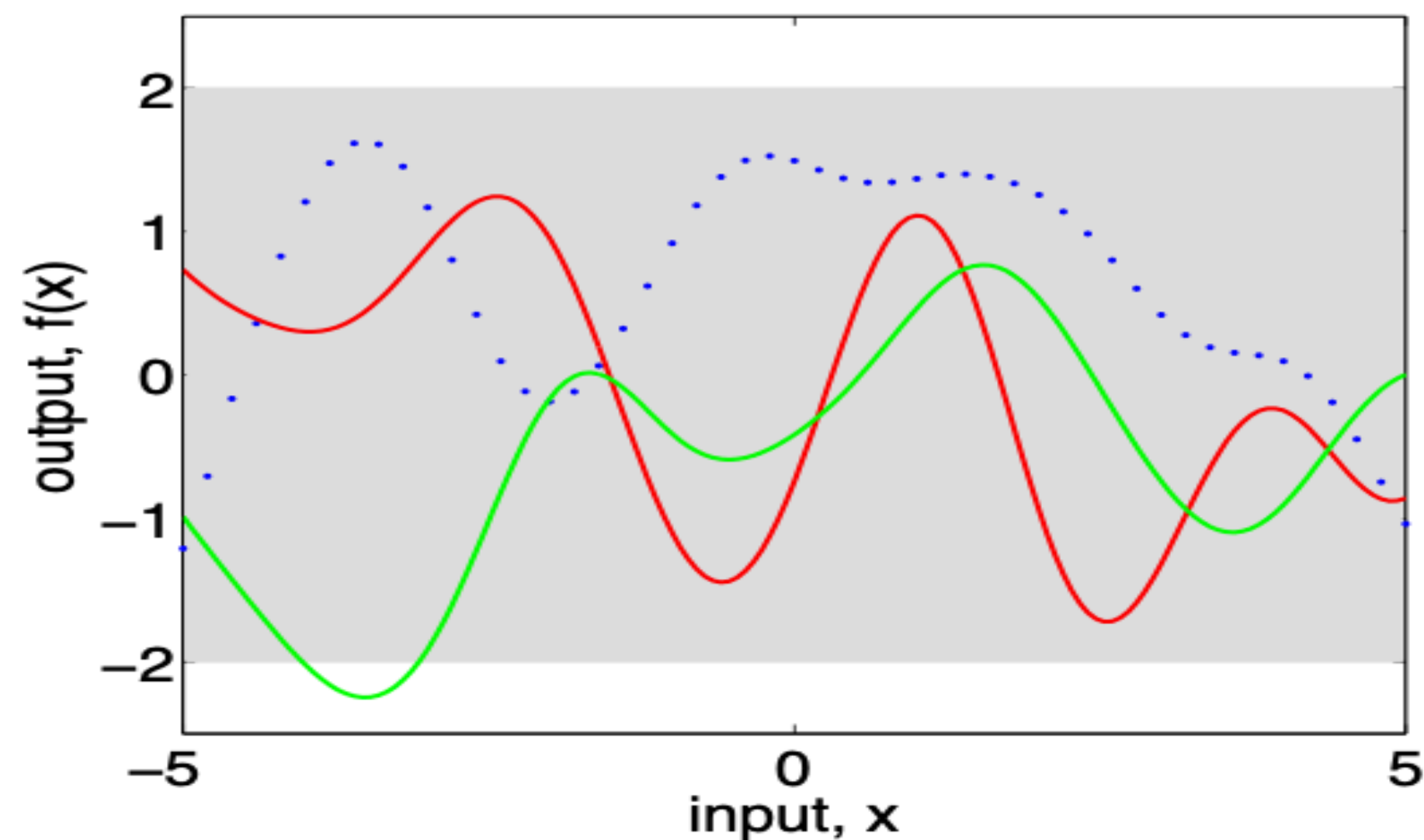
Gaussian Process Regression

Datasci 224 project

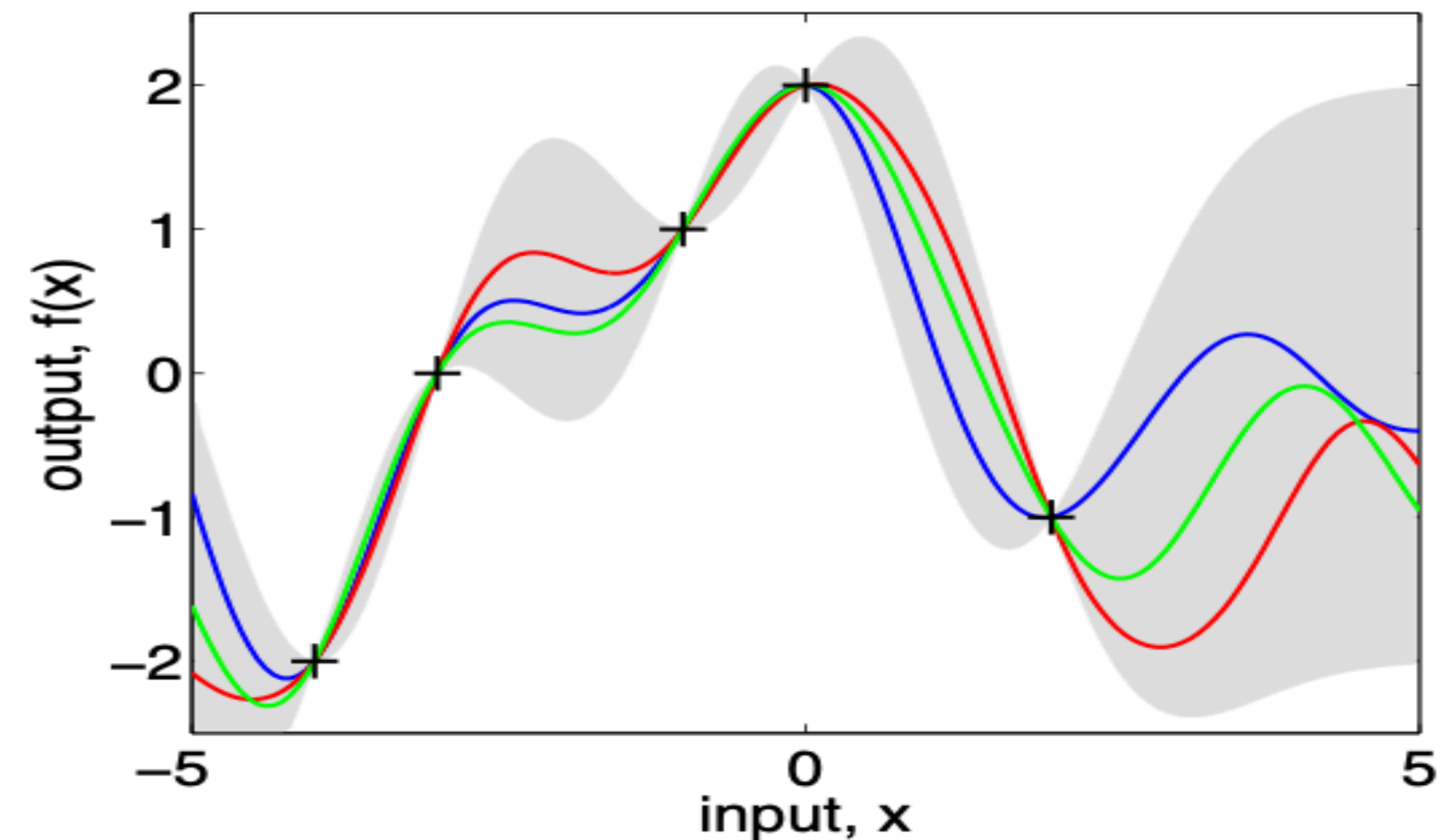
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Overview

- Objective: to develop a data driven algorithm to maintain smoothness/continuous progression across the domain without pre-specifying parameters (in contrast to standard regression)
- GPR is a non-parametric algorithm, bayesian approach to regression



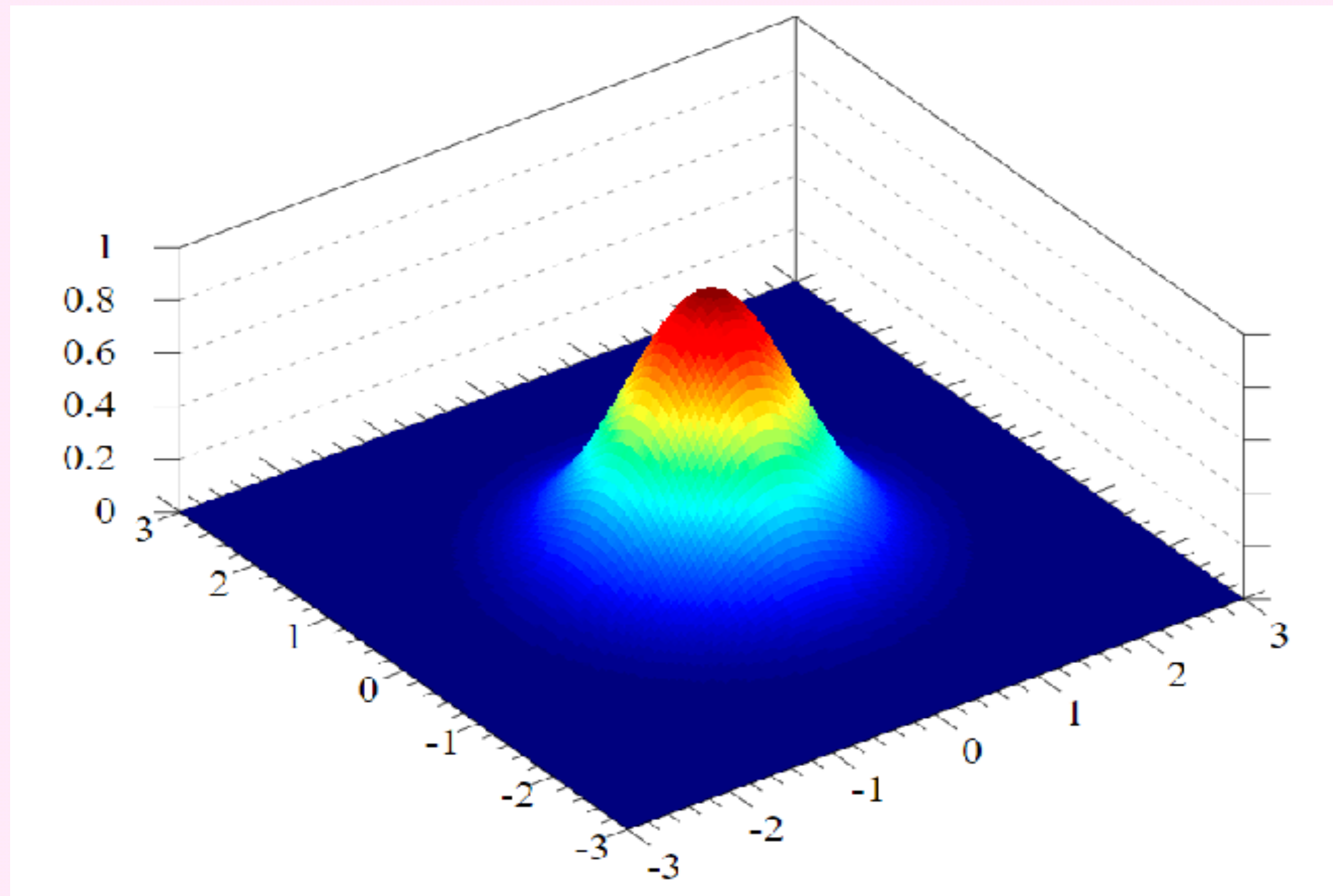
(a), prior



(b), posterior

Visualizing

- So how do we define smoothness for the algorithm?



Every point in the continuous space takes influence / influences its surrounding space following a covariance (kernel) function

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')).$$

Algorithm

Example: RBF kernel

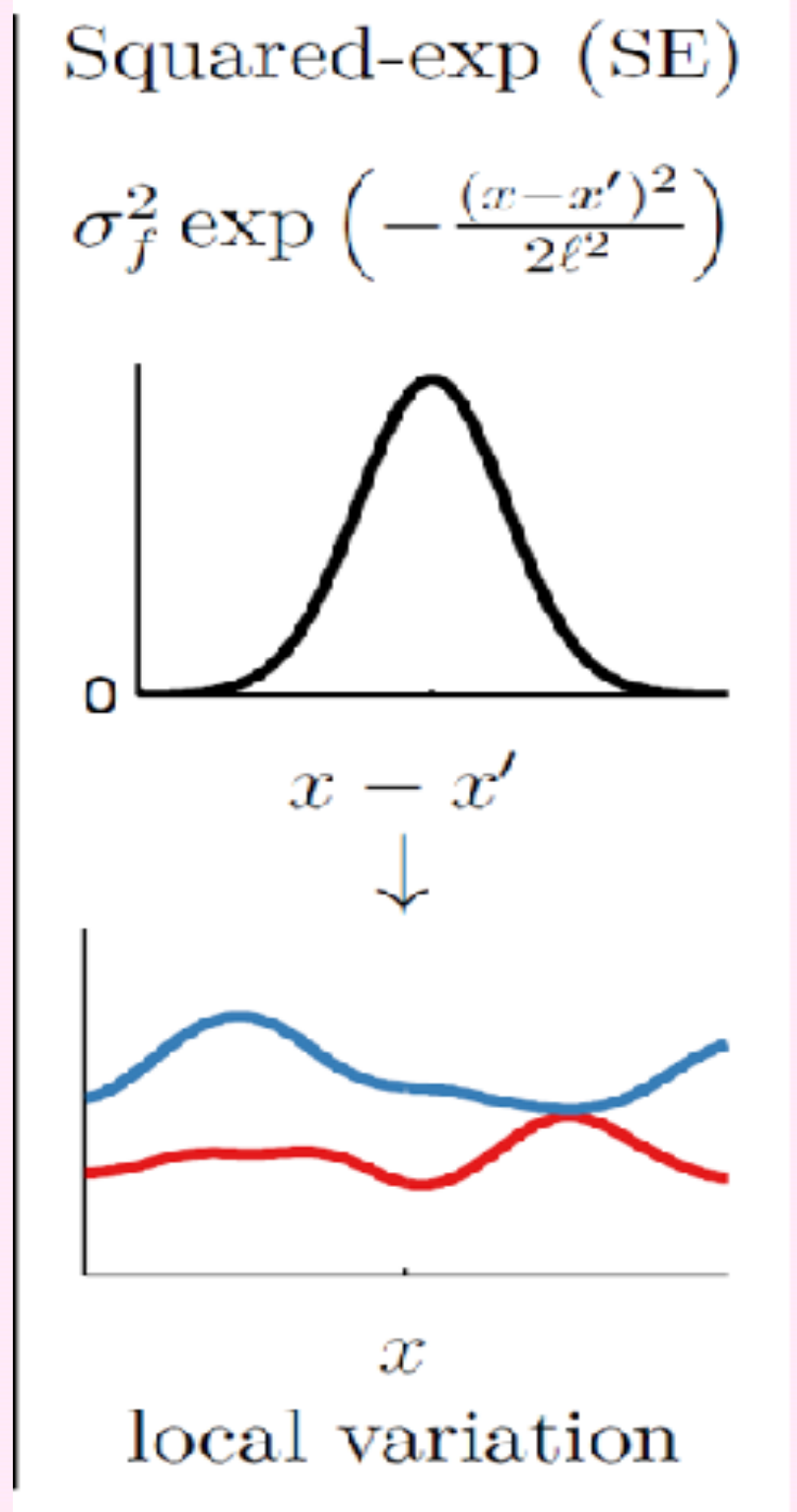
$$\underbrace{k_y(x_p, x_q)}_{\text{Covariance Kernel}} = \sigma_f^2 \exp\left(-\frac{1}{2\ell^2}(x_p - x_q)^2\right) + \sigma_n^2 \delta_{pq}.$$

↑
Covariance Kernel

- ℓ : the “standard deviation” of the rbf kernel
- σ_f^2 : signal variance
- σ_n^2 : noise variance

For every pair of points in the continuous space

} Hyperparameters

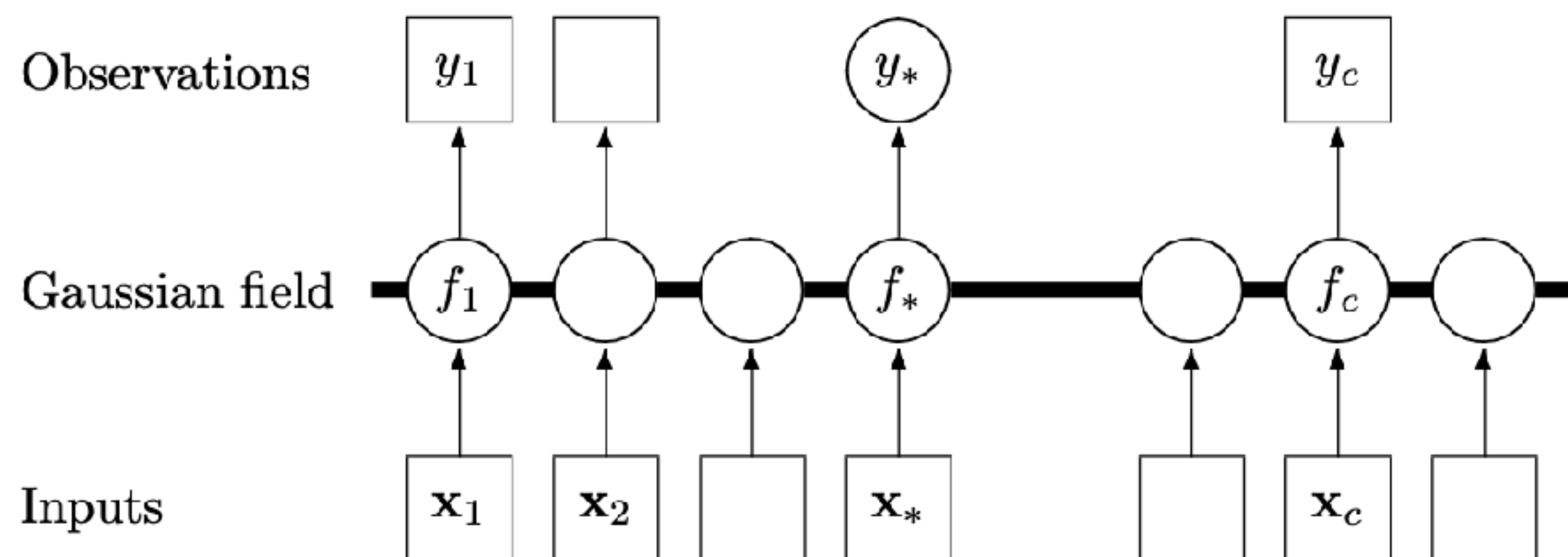


How we optimize (hyperparameter tuning)

- Similar to other algorithms,
 - we can define penalties to penalize extreme fluctuations for example
 - We can also iterate over a grid of hyperparameters

Algorithm

- The algorithm transforms the training outcome through the inverse covariance matrix to calculate a set of weights which corresponds to posterior mean
- Then it uses the covariance matrix between the test data and training data to combine those weights to reach a final prediction

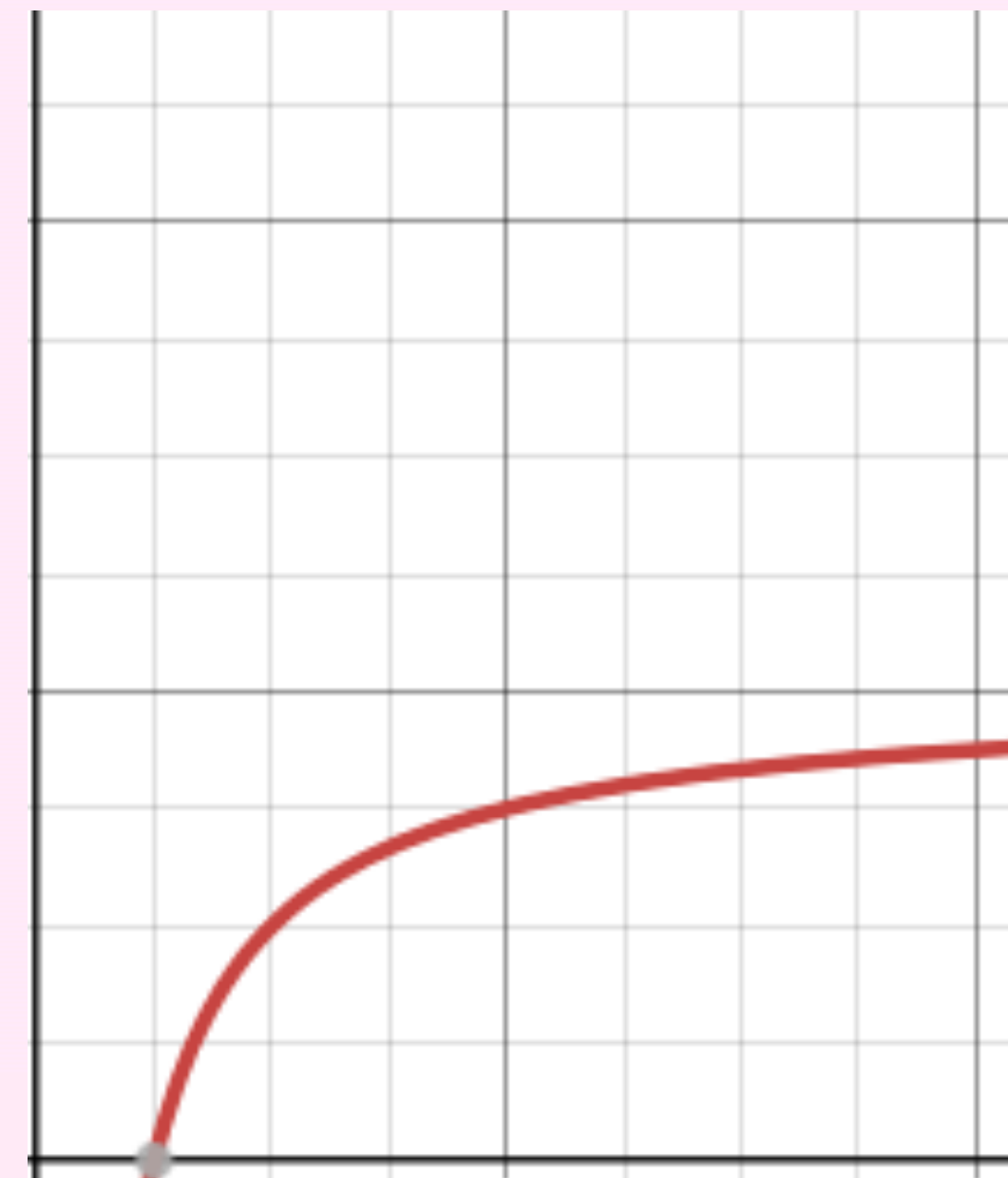
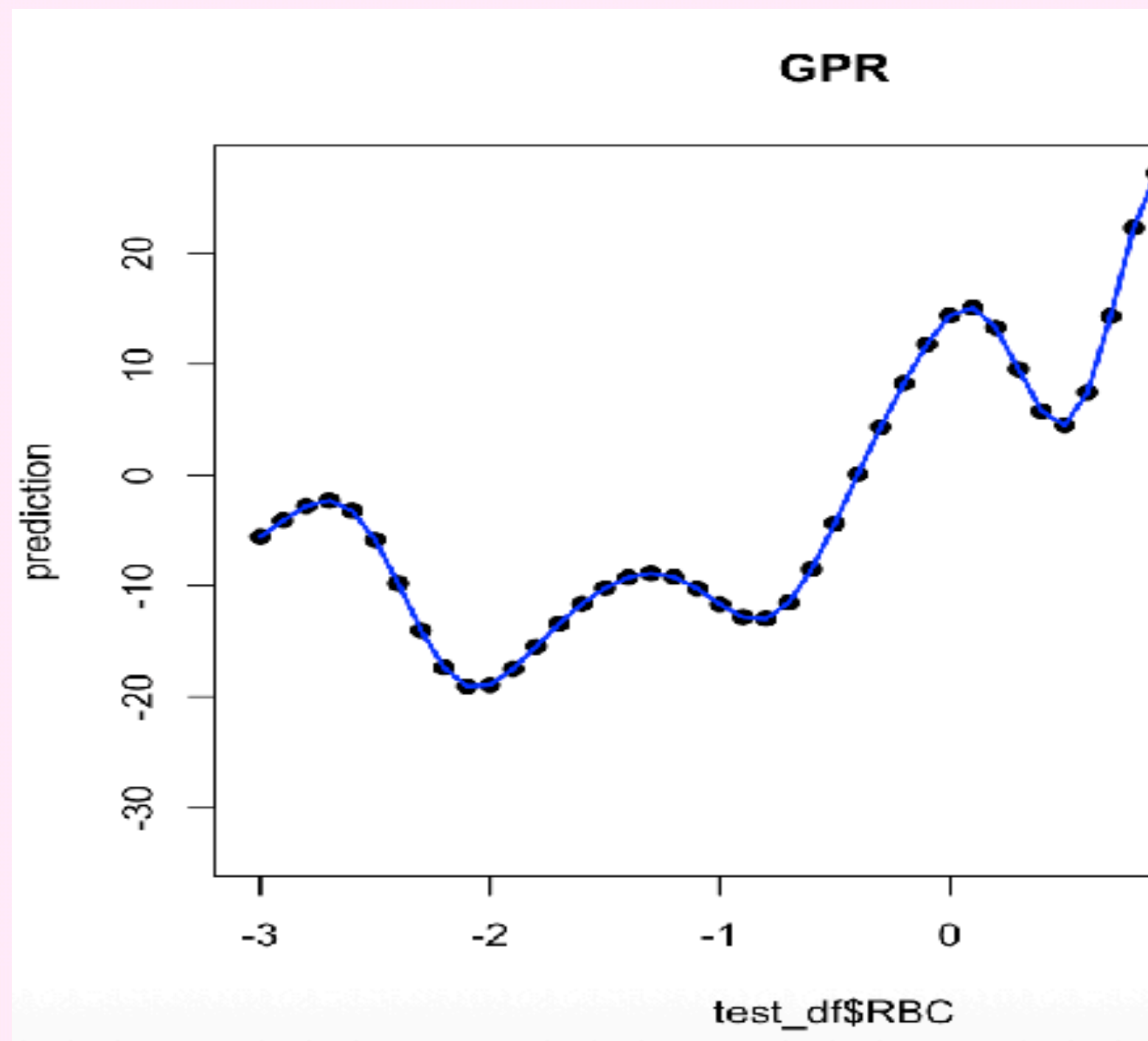


Applying to data

- For my problem statement, I had a system with a perceived mathematical relationship based entirely on biological knowledge
- We want to assess/confirm our perceptions
- Example: $X_{t+1}/X_t = r_1(1 - \frac{X_t + \alpha Y}{A})$
- We can use a fabricated test data to hold all variables constant except one, for example, to visualize the relationship

Applying to data

- $X_{t+1} = r_1(1 - \frac{X_t + \alpha Y}{A})$
- A is RBC(Red Blood Cell) and I want to assess this variable's behaviour across its domain
Perceived



Applying to data

- Can also skip the mathematical relationship and create a data driven temporal model
- Example model X_t in terms of t (days) and the initial profile

**Thank you for listening
Questions**